

# ALTOS® Lite Loose Tube, Gel-Free, Single-Jacket, Single-Armored Cable 144 F, SMF-28® ULL fiber, Single-mode (OS2)

CORNING

**Part Number:**  
**144PUC-T4119D20**

Corning ALTOS® Lite gel-free, single-jacket, single-armored cables are designed for campus backbones in direct-buried installations. The loose tube design provides stable and highly reliable transmission parameters for a variety of voice, data, video and imaging applications. These cables also provide high-fiber density within a given cable diameter while allowing flexibility to suit many system configurations. The single armored construction provides additional crush and rodent protection with a high-strength ripcord under the armor for easy stripping. Gel-free means the cables are fully waterblocked using craft-friendly, water-swappable materials which make cable access simple and require no clean up. The flexible, craft-friendly buffer tubes are easy to route in closures, and the SZ-stranded, loose tube design isolates fibers from installation and environmental rigors while allowing easy mid-span access. These cables have a medium density polyethylene jacket that is rugged, durable and easy to strip.

## Features and Benefits

### Gel-free waterblocking technology

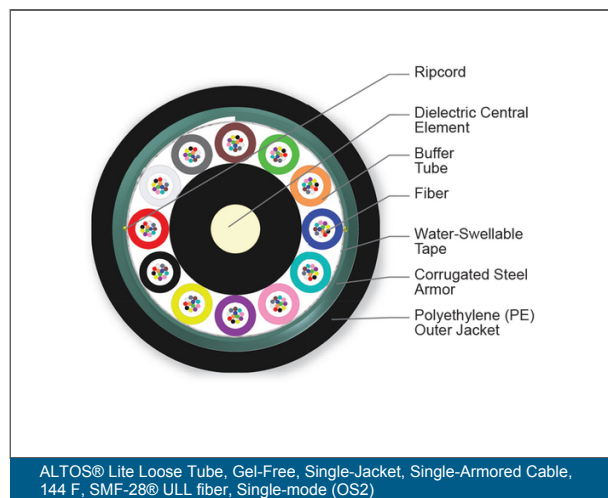
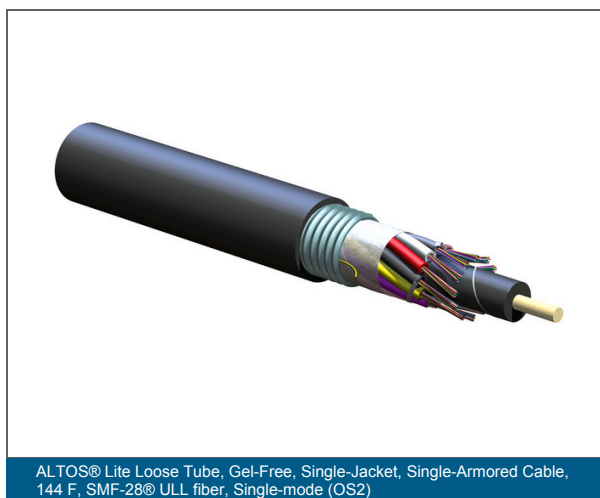
Craft-friendly cable preparation

### Polyethylene jacket

Rugged, durable and easy to strip (while providing superior protection against UV radiation, fungus, abrasion and other environmental factors)

### Corrugated steel tape armor

Provides rodent resistance for direct-buried applications



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## Specifications

### General Specifications

|                      |                                   |
|----------------------|-----------------------------------|
| Installation Methods | Aerial, Direct Buried, Duct       |
| Cable Type           | Loose Tube                        |
| Environment          | Outdoor                           |
| Product Type         | Steel Tape Armor                  |
| Fiber Category       | Ultra Low-Loss, Single-mode (OS2) |
| Cable geometry       | Round                             |

### Standards

|                          |                                                                                                                                |
|--------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| RoHS                     | Free of hazardous substances according to RoHS 2011/65/EU                                                                      |
| Common Installations     | Outdoor lashed aerial, duct and direct-buried, indoor when installed according to National Electrical Code® (NEC®) Article 770 |
| Design and Test Criteria | ANSI/ICEA S-87-640                                                                                                             |

### Environmental Conditions

|                                 |                                                                                                                                                                                                                |
|---------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Notes                           | Corning recommends storing cable in a proper temperature environment prior to installation to allow the cable temperature to meet installation temperature range specifications for best installation results. |
| Temperature Range, Installation | -30 °C to 70 °C (-22 °F to 158 °F )                                                                                                                                                                            |
| Temperature Range, Operation    | -40 °C to 70 °C (-40 °F to 158 °F )                                                                                                                                                                            |
| Temperature Range, Storage      | -40 °C to 70 °C (-40 °F to 158 °F )                                                                                                                                                                            |

### Cable Design

|                    |            |
|--------------------|------------|
| Central Element    | Dielectric |
| Fiber Count        | 144        |
| Number of Ripcords | 2          |
| Outer Jacket Color | Black      |

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## Cable Design

|                                                     |                                                                                  |
|-----------------------------------------------------|----------------------------------------------------------------------------------|
| Outer Jacket Material                               | Polyethylene (PE)                                                                |
| Tensile Strength Elements and/or Armoring - Layer 1 | Corrugated steel tape armor                                                      |
| Buffer Tube Color                                   | Blue, Orange, Green, Brown, Slate, White, Red, Black, Yellow, Violet, Rose, Aqua |
| Buffer Tube Diameter                                | 2.5 mm (0.1 in)                                                                  |
| Number of Active Tubes                              | 12                                                                               |
| Number of Tube Positions                            | 12                                                                               |
| Tape                                                | Water-swellaable                                                                 |
| Fiber Coloring                                      | Blue, Orange, Green, Brown, Slate, White, Red, Black, Yellow, Violet, Rose, Aqua |
| Fibers per Tube                                     | 12                                                                               |
| Color Code Standards                                | Telcordia                                                                        |

## Mechanical Specifications

|                                   |                     |
|-----------------------------------|---------------------|
| Max. Tensile Strength, Long-Term  | 890 N (200.08 lbf)  |
| Max. Tensile Strength, Short-Term | 2700 N (606.98 lbf) |
| Nominal Outer Diameter            | 17.5 mm (0.69 in )  |
| Min. Bend Diameter Installation   | 526 mm (20.71 in)   |
| Min. Bend Diameter Operation      | 350 mm (13.78 in)   |

## Optical Characteristics

|                         |                             |
|-------------------------|-----------------------------|
| Fiber Code              | P                           |
| Performance Option Code | 19                          |
| Fiber Category          | OS2                         |
| Fiber Type              | Single-mode (OS2) / 250 µm  |
| Fiber Name              | SMF-28® ULL                 |
| Maximum Attenuation     | 0.33 dB/km / - / 0.19 dB/km |
| Wavelengths             | 1310 nm / 1383 nm / 1550 nm |
| Fiber Compliance        | ITU-T G.654.C               |

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| Dimensions   |                               |
|--------------|-------------------------------|
| Cable Weight | 245 kg/km (164.63 lb/1000 ft) |



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